

Recommender System of Final Project Topic Using Rule-based and Machine Learning Techniques

Cut Fiarni

Department of Information System
Institut Teknologi Harapan Bangsa
Bandung, Indonesia
cutfiarni@ithb.ac.id

Herastia Maharani

Department of Information System
Institut Teknologi Harapan Bangsa
Bandung, Indonesia
Herastia@ithb.ac.id

Billy Lukito

Department of Information System
Institut Teknologi Harapan Bangsa
Bandung, Indonesia
Blukito@ithb.ac.id

Abstract—The final project is a mandatory graduation requirement for bachelor's degree students. However, students often having problems in determining topic that is suitable with their interests and competencies. As a result, some students might have to change their topics halfway, which can affect their study period. Ironically, the abundant volume of previous final project documents available in the university library only add more confusion and difficulty for the students in finding relevant references for their research topic. Therefore, the focus of this research is to implement a machine learning approach to analyze and model an algorithm to recommend final project topics, based on student's interest, competencies, and their respective supervisor. This research also aims to establish a framework to map academic attributes, as part of feature selection. As the result, we develop a recommender system based on cosine similarity algorithm to recommend topics based on similarity between student's profile and topics represented by lists of keywords. Performance is measured by comparing the recommendations given by the proposed system against the actual topic chosen by students, with a very satisfying result of 71.43% precision.

Keywords—final project, machine learning, feature selection recommender system, cosine similarity

I. INTRODUCTION

Final Project is one of the obligatory subjects for undergraduate students in all universities to obtain their bachelor's degree. The final project, thesis, or dissertation is a form of research that had to be done by students through certain standards and processes [1]. However, it is widely known that final project has the potential to become a burden for students. In worst case scenarios, the burden of final project could even cause the students to graduate late or even dropped out of college. In composing the final project, research is carried out as a mandatory step in solving problems raised in the topic of the final project. Moreover, topic of final project usually based on student concentration or sub major, a continuation of previous research problem, by implementing different methods or broaden the research approach and frameworks. According to previous research, most of student think that composing final project is a scourge [2]. Generally, these hindrance related to their motivation and willingness to formulates their research problems, then to analyze and formulate the alternative solution. Thus, it can be concluded that the root of the problem is in determining the appropriate the research topic.

The university library that collected years of final project document, contain a wealth of hidden knowledge that could help students in their research. Ironically, this increasing volume of documents with a variety of problems and method solutions, only add the confusion and difficulty for students to choose which documents are relevant as references for their research topic. These document are large in volume, variety, and value of data, and thus need to be analysed effectively in

order to obtain the information or patterns contained in it. One method that is quite capable of analysing data is machine learning, namely a data extraction method which is a combination of artificial intelligence and computer science [3].

The focus of this research for the field of information systems. The academic curriculum of the information systems encompasses information technology, management of information system, information systems development, and business processes of organizations [1]. The research is conducted on department of Information System (IS) at Institut Teknologi Harapan Bangsa (ITHB) located in Bandung, Indonesia. Because of its broad and complex study area, IS department divided its specialty into three sub majors, which are Infrastructure Management, Business Intelligence and E-Business [4]. Based on the questionnaire conducted on student and alumni of the IS department, 20% of respondents experienced obstacles in determining final project topics that were in accordance with their interests. Furthermore, 33.3% of respondents said that it was difficult to find previous final project references that are suitable with their chosen topic. In searching for topics that are relevant and matched with preferences, readers also want indexers to get relevant documents easily and accurately [5]. Moreover, 53% of students have changed topic and potentially waste one month of their research time, while 77% of respondents proposed topics of their final project only one week before the deadline for supervisor's approval which obviously delay their research progress.

The objective of this research is to implement machine learning technique to build a profile for each student based on their interests and competencies, and then give personalized recommendation of relevant references for final project topics. This personalized system is called a final project recommender system (RS). This system would help student to cope with information overload by providing them with relevant items of information. Student's interest and competencies can be modelled from their sub-major, grades, or even job interest that each can be associated with certain topics and respective supervisor. On the other hand, courses and previous final project references also have association with certain topics and the supervisor's areas of expertise. A simulation would make the educational process more interactive, which not only make the student gain knowledge they needed and also understand their force that influence it [6]. The Personalized feature could help students gain the recommendation of their final project topic, supervisor, and references from previous work.

II. RELATED WORK

Research of machine learning in education field is an emerging research approach in term to how generated hidden knowledge of academic data driven to learn and model a better

understanding of the current process and to build technology as solution for of education problems. The basic principle of machine learning is to utilize data to create statistical models, these models are generally used by the system to make future predictions based on past input data or learn hidden patterns contained in the data. One of the main advantages of machine learning is the ability to modify and adapt in response to changes in data [7]. Previous related research of machine learning in education are to predict student performance using NB Tree [8]. Another research using supervised machine learning, C4.5 decision tree algorithm to extract hidden pattern for the selection of IS sub majors, in accordance with the interests and competencies. These hidden patterns use as model to build academic decision support system[4]. Another research compares the performance of machine learning algorithms to predict number of students take the course. In this study, predictions are modelled based on the attributes of student data, namely Grades, GPA, Cumulative GPA, Semester Credits, Cumulative Semester Credits and Semester [9]. Another study offers a selection subject recommendation system that builds on the profile data and student's interest using the technique of Association Rule [10].

Recommender systems (RS) are broadly utilized to assist users in handling the great volume and variety of information, particularly in searching for the most appropriate content tailored to the specific preferences of the user. Providing knowledge-base embedding (KBE) as model of personalized recommender system is crucial for highly accurate performance. Moreover, integrating large-scale structured data, while adopting collaborative filtering makes it possible to learn user and item representations while preserving the structure of their relationship with external knowledge for explanation [11].

Model component of Personalized Recommender System are classified into four categories, which are collaborative filtering, content based, demographic and hybrid. Collaborative filtering builds recommendations by learning from user-item historical interactions. Content-based recommendation is based primarily on comparisons across items' and users' auxiliary information, such as texts, images and videos. Demographic algorithm learn from personal information of user (such as age, gender, location, etc), and then it categorized based on the demographic attribute. Hybrid model refers to recommender system that integrates two or more types of recommendation strategies [12]. The Cosine similarity method proofed surpasses the performance of Euclidean Distance and MoreLikeThis from Apache Lucene; and as such is suitable to implement on the recommender system [13]. In general, a similarity function is a function that accepts input of two objects then calculates similarity between the two objects and returns in a form of real numbers. The value returned by the similarity function generally ranges in interval $[0..1]$. The greater the similarity function result of the two objects evaluated, the more they are considered to be similar. In a function that produces a value in a range $[0..1]$, value of 1 means that two objects are exactly the same, while a value of 0 means the opposite [14].

In this research, the proposed system can deal with the problem of information overload related to topic selection, as well as references of previous final project documents as items. The objective is to build machine learning model to recommend the most relevant topics and final project documents based on the predicted interests and abilities of

each student, based on attributes related to information about final project topic, users and the relationship between topic and users. The method that will be used to get a solution to the existing problems is content based filtering method. This method is relevant to human behavior of the matching process between student academic performance and job interest with the description or characteristics of topic.

III. METHOD

A. Framework

In this section we will explain the method framework as guidance to feature selection process as part of model developing process of the proposed system. Fig. 1 describes our proposed framework of research method which consists of two main processes, mapping data and calculating the recommendation. The goal of the first part of this research is to map the relationship between user preference and final project topic items, which would be represented in the form of a matrix. This matrix would be used in the second part of this research, which are calculating the cosine similarity value, and then compared to each other in order to gain the top 5 highest value].

B. Feature Selection and Data Pre-processing

In this data driven machine learning process, we used primary and secondary data. The primary data is collected from final project documents, list of final project research topic from each of IS department lecturer, and IS department course syllabus. The secondary data is collected from previous research that consist of mapping IS sub major in the job field, courses and its acceptable grade level [4]. ITHB Academic Guideline Book will be used to gain keyword for each of the 57 courses in information system department. The objective of this step is to prepare the collected dataset and analyses its attribute as a rule-based feature selection technique.

Feature selection is an important technique for extracting the most essential feature of the data set that correlates with the machine learning algorithm of system solution. The feature selection techniques would enhance the performance system in several aspects like accuracy and simplicity, and more importantly it reduces the dimensionality and noise from dataset with multiple characteristic such as document of final project. Therefore, the abstracts of final project documents are transformed from the full text version to vectors which describes the contents.

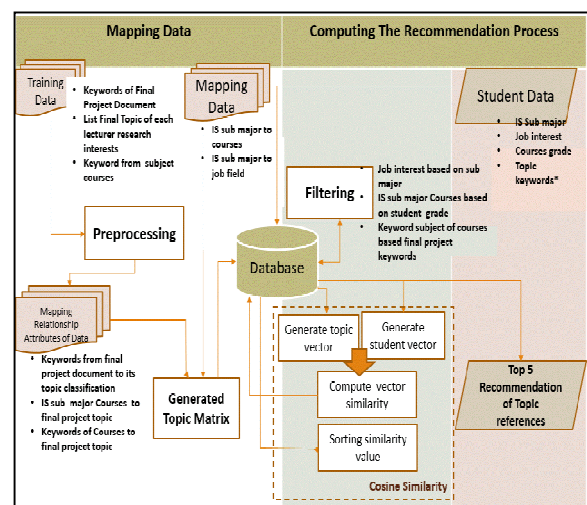


Fig. 1. Research Framework

In the vector space model of Final Project Topic, documents are represented as vectors of features representing the characters of documents and its data driven academic value regarding trend of research in information system. Vector Space representation of a document involves four steps. First is document indexing, where content bearing keywords are extracted from the documents, and also map the keywords and terms from dataset. The second step is to compute the weights of indexed keywords to enhance retrieval of documents relevant to the user competency and interest. This step involves pre-processing as shown on Fig 2.

Pre-processing consists of attribute selection, cleaning, and normalization. In this step, the attribute data that have no value for this research would be removed. We also re-write keywords that have the same meaning, so there would be no redundancies accumulates. A relationship map is created along with the keywords of each Final Project subjects. Table 1 shows the significant attributes of each of the primary and secondary data and the result of mapping relationship of attributes of topic and student.

The next steps of vector space representation is identifying the similarities between the matrix of user competency and interest and matrix of final project topic. The final step is identifying the similarities between the documents. The next sub section would describe these last two steps.

C. Recommendation Technique

In this research, we compare between the three most popular algorithms for recommender system, which are Keyword Matching (KM), Nearest Neighbour (NN) and cosine similarity (CS) method. KM algorithm work by finding matches between the content of item with user profile. The result of this method is a binary value, match or unmatched (1/0). Meanwhile, in NN algorithm, both item and user profile will be formed into the matrix and then the distance between the user profile matrixes will be calculated. The shorter the distance between the matrices, the higher the level of similarity of the user profile and the items. The result of this algorithm is range value 0 to infinity [15].

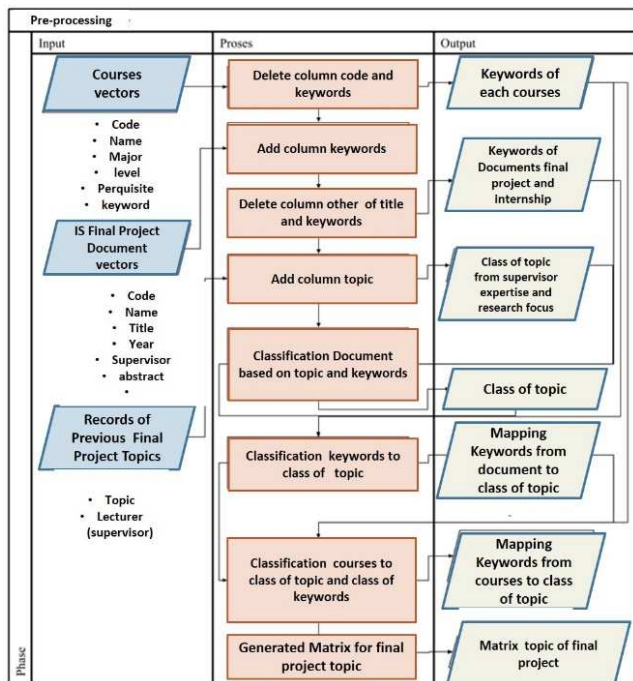


Fig. 2. Pre-processing framework

TABLE I. ATTRIBUTES OF FEATURE SELECTION

Dataset	Total	Type of Data	Attribute					
			FP Title	Keyword	Topic	Lecturer	courses	Sub major
Final project documents	276	Primary	V	V				
List of research topic	32				V	V		
Course's syllabus	57			V			V	
Mapping course to IS sub major		Secondary					V	V
Mapping job field to IS sub major							V	V
IS course to final project topic		Mapping Relationship			V		V	V
final project document to its topic classification			V	V	V	V		
Keywords of subject courses to final project topic	425			V	V		V	

Since the characteristics of the data to be calculated has only one criterion, which are the keyword for the user, and the nearest neighbour algorithm uses a matrix, that requires more than one criterion. As shown on table II, since the characteristics of the data to be calculated only have one criterion, which is keywords, and NN algorithm uses a matrix that requires more than one criterion, the NN does not match the characteristics of the data to be calculated. Furthermore, it can be concluded that both NN and KM algorithm does not match the characteristics of the model needed for this proposed system. Similarities between object can also be measured using other methods such as clustering as shown in [16], but this approach is not suitable for non-numerical data like keywords.

Therefore this research focuses on the design of optimization of CS methods for educational applications that are used to generate recommenders. As shown on Fig 1, there are two main step build the system. In the first step, the modified content-based filtering methods is finding similarity between final project documents with user personalization preferences that were extracted from student jobs interest, their grade of courses related to their sub-majors category, and also from their internship project. In the second method, the specific feature-based customized recommender works in the training and recommendation steps by generating topic matrix and filtering strategy based on user personal preference datasets. In this step a matrix is generated following these steps: use all keywords from final project documents as rows, use all class of topic as columns, fill each field with 1 if the keyword is relevant to the topic, or with 0 if otherwise.

Table II illustrate the generated matrix that shows the relationship of each keyword to the class topic and will be used as the calculation model by the proposed system. As the result from 276 documents of previous final projects used in data processing, we obtained 425 keywords and 32 topics producing a 425x32 matrix.

TABLE II. FINAL PROJECT TOPIC MATRIX

FP keyword/Topic	knowledge management system	IT governance	Business simulation	..	Topic -n
repository	1	0	0	..	
keyword-n	...	...	...	..	

The user profile vectors are built from their grades for related IS sub major courses, and keywords obtained from mapping their job interest to IS sub major. The keywords are identical with the keywords used in the final project topic matrix. Thus, each student will be represented by a vector of 425 elements. The default value of this vector is 0, when there's a match between a keyword from final project matrix and keywords from the student's sub-major or job interests then the value is changed to '1', an example of the user vector is [0,1,0,0....].

The user vector is then compared with each topic vectors from topic selection matrix using cosine similarity. The resulting similarity ranges from -1 meaning exactly opposite, to 1 meaning the same, with 0 indicating orthogonality or decorrelation, while in-between values indicate intermediate similarity or dissimilarity [17]. Given two vectors of attributes, A and B, the cosine similarity, $\cos(\theta)$, is represented using a dot product and magnitude as [18]

$$\text{Similarity} = \cos \theta = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (1)$$

IV. RESULT AND DISCUSSION

The algorithm of proposed system and experimental result is discussed in this section. The experiment will clarify the rule-based attribute extraction of student profile and the algorithm of the recommendation of final project topic and reference. The system performance is measured through number of experiments of proposed system against human categorization behavior.

A. Algorithm of proposed system

The algorithm of proposed system illustrated on Fig 3. The following are several steps and data used during the filtering phase and possible results that will be formed:

- Initial Input. The system started with initial input from user that consist of student identity, job field interest and their chosen IS sub major.
- Filtering courses based on sub major. Mapping the relationship of these attributes is used to filter which courses are correlated with student interests.
- Input student grade for filtering courses result. In this step, the corresponding grade for the filtered courses is processed to gain information about student competencies.
- Input initial interest for topic keywords (optional)
- Generate user vector and topic vector. By using the list of student keywords that have been input in the previous stage and list of keywords of the final project report, a user vector will be generated. For one topic selection vector, it has total 425 keyword values that work incrementally, so it can still be added if there are new additional keywords

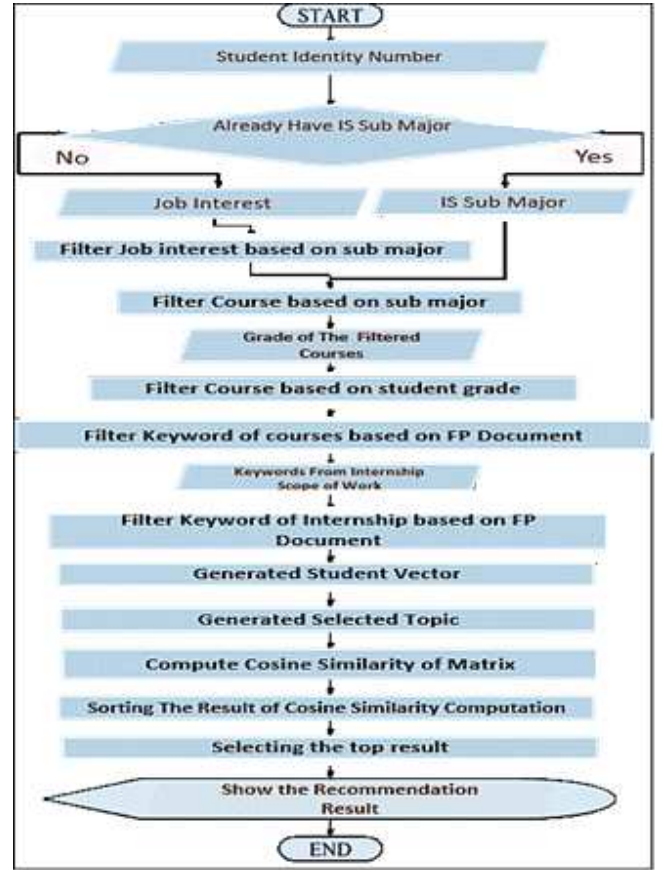


Fig. 3. Algorithm of The Proposed System

- Calculate cosine similarity. After the topic selection vector is formed, the similarity of each topic selection vector and user vector is calculated [18].
- Give the top 5 highest recommendation. From the result of the cosine similarity calculation, rank the score from the largest to the smallest and select only the five topics with highest score. The system then recommends the top five topics that are most similar with the user profile

B. Design and Development

The Unified Modelling Language (UML) is used to explain the flow of the system that will be developed. One of the diagrams that will be used in the design of this system are use case diagrams. Use Case Diagrams used to find out the functions that exist in the information system and who has the right to use the functions. There are three type of user for the proposed system as shown on Fig 4. Students can get recommendations by directly accessing the main page on the system without login. This main home page contains information about how the system works, and give recommendation based on the student data.

Fig. 5 shows the recommendation results displaying the five results of topic recommendations that are most suitable for students, the results of calculations, previous final project references, and detailed input data that students have done in the previous stage. Fig 6 shows the system's dashboard, that displays the core information of the system, which are: the number of recommendations that have been given, the number of keywords registered in the system, the number of final project research reports registered in the system, the number of topic choices registered in the system, the number of recommendation results that matches the user, five most

frequently recommended topics, five most frequently used keywords, five most frequently submitted job fields, the total number of sub-majors inserted and the top five topics with the most reports registered in the system.

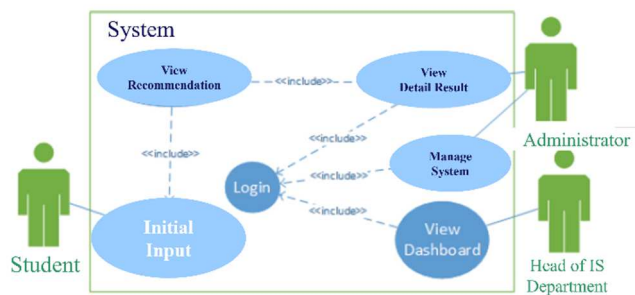


Fig. 4. Use Case of the recommender system.

Recommendations		
Topics	Calculation Results	Suggested References
knowledge management system	0.324497	
data mining	0.294174	
data warehouse	0.154503	
system informasi media	0.138675	
decision support system	0.118864	
Input Detail		
NIM		
Job Interest	Computer Programming	
Sub Major	Computer System Analyst	
	B - Business Intelligence	: B
Transcript	Mathematical Business I	: B
	Data Management and Information	: A
	Process Business Analysis	: B

Fig. 5. Recommendation Page

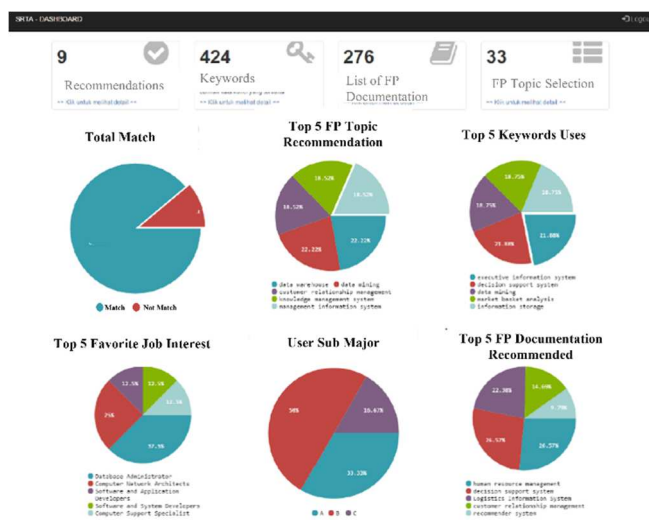


Fig. 6. Dashboard

C. Performance Evaluation

To evaluate the performance of the recommendation system is relevancy assessment. Precision is defined as the proportion of relevant items amongst retrieved items. Precision is also related to the system's ability not to call irrelevant documents [18].

Objective performance measures reflect the model generalizability and are the results of cumulative classification successes and failures of the test set, which are given in a confusion matrix [19]. The topic keywords is compared to the topic that students have taken, along with feedback provided

by the student on whether they found any suitable topics from recommendations given by the system or not. The precision of this system uses training data in the form of data students who are taking the final project, which is the basis in data processing to get an accurate system.

$$\text{Precision} = \frac{\text{Number Relevant Document Retrieve}}{\text{Number of all Document Retrieved}} \times 100\% \quad (2)$$

The proposed recommendation system performance were being evaluated using students data from the academic year 2015/2016, resulting in a precision value of 71.43%. Precision is obtained from the number of FP topic recommendations based on interests and competencies, and then it matched to the actual topic of the test data. The resulting precision shows that the Final project Topic Selection Recommendation System Using Content Based Filtering Method is quite accurate. Findings show that using cosine similarity and filtering algorithm is effective to simulate a recommending topic of student's final project based on their interest, competency and preference.

V. CONCLUSION

This research focus on implementing educational data mining to solve common problem of the university student in choosing the topic for their Final Project research, suitable with their personalized interests and competencies. This research modeled and designed the Final Project Topic Selection Recommendation System, by implementing information retrieval and data mining. A total of 276 TA report records were used in data processing to produce system rules in the form of a 425x32 matrix, representing 425 keywords and 32 topics. The rules used in this recommendation system are incremental, in which they will dynamically increase or change according to the addition of master data by the administrator. From the system performance evaluation, we conclude that Content-based filtering method with the cosine similarity algorithm is the most suitable in providing recommendations, with the precision value of 71.43%, showing that the proposed system could be implemented and helped solve the problem. Moreover, the dashboard provided in the system could help Head of Department to make better decision regarding problems that could be encountered by their student. For the future work, integration of the proposed system with the ITHB Academic Information System, especially to import value form their transcript, could improve the efficiency of the system. Features feedback or reputation in recommendations can be integrated into the proposed system with intelligent optimization of the algorithm, so the system could learn from the computation process and give better recommendation.

REFERENCES

- [1] Ministry of Education and Culture, The regulation minister of Education and Culture of the Republic of Indonesia. Article Number 3, concerning the national standards of college, article 49 paragraph 5, 2020
- [2] L. Malika, "Kecenderungan Pemilihan Topik dalam Penyusunan Judul Skripsi Mahasiswa Program Studi PG PAUD Tahun 2010 – 2015" (final project) Semarang : Universitas negeri Semarang, 2016
- [3] A. C. Müller and S. Guido, "Introduction to ML with Python: A Guide for Data Scientists". California: O'Reilly Media, Inc, 2016
- [4] C. Fiarni, E. M. Sipayung, and P. B. T. Tumundo., "Academic Decision Support System for Choosing Information Systems Sub Majors Programs using Decision Tree Algorithm," in J. Inf. Syst. Eng. Bus. Intell., vol. 5, no. 1, 2019, p. 57.

- [5] M. I. Subroto, T. Sutikno, and D. Stiawan, "The Architecture of Indonesian Publication Index: A Major Indonesian Academic Database," in TELKOMNIKA (Telecommunication Computing Electronics and Control), vol. 12, 2014. p. 6
- [6] A. Gunawan, C. Fiarni, and G. Lawalata., "Business process learning system with real-time simulation approach." Paper presented at the 2nd International Conference on Human Capital and Knowledge Management (ICHCKM 2015), Indonesia, Bandung, 2015
- [7] S. Marsland, Machine Learning: An Algorithmic Perspective. USA: CRC Press, 2015
- [8] T.M. Christian and M. Ayub, "Exploration of Classification Using NBTree for Predicting Students' Performance," International Conference on Data and Software Engineering, 2014.
- [9] F.A.Bachtiar, K. Indra, Syahputra, and S.A. Wicaksono, "*Perbandingan Algoritme Machine Learning untuk Memprediksi Pengambil Matakuliah*", in JTIK(Jurnal Teknologi Informasi dan Ilmu Komputer), Vol.6, No. 5, 2019, p. 543-548
- [10] D.P. Kusumaningrum, N.A. Setiyanto, E.Y. Hidayat, and K. Hastuti, "Recommendation System for Major University Determination Based on Student's Profile and Interest," n Journal of Applied Intelligent System, 2(1), 2017, p 21–28
- [11] Q. Ai, V. Azizi, X. Chen, Y. Zhang, "Learning heterogeneous knowledge base embeddings for explainable recommendation. Algorithms", MDPI, 2018, vol 11, p. 137
- [12] Dietmar Jannach, Markus Zanker, Alexander Felfernig, and Gerhard Friedrich., Recommender systems: an introduction. 2010
- [13] N.A.Rakhmawati, A.A.Firmansyah,P.M.Effendi, R.Abdillah and T.A.Cahyono., "Auto Halal Detection Products Based on Euclidian Distance and Cosine Similarity, " in IJASEIT, Vol.8 No. 4-2, 2018, pp 1705-1711.
- [14] C. Fiarni, H. Maharani and N.Calista, "Product Recommendation System Design Using Cosine Similarity and Content-based Filtering Methods," in IJITEE, Vol 3, No 2, June 2019, pp 42-48.
- [15] Teng S., Li J., Li R., Zhang W. (2014) The Calculation of Similarity and Its Application in Data Mining. In: Zu Q., Vargas-Vera M., Hu B. (eds) Pervasive Computing and the Networked World. ICPCA/SWS 2013. Lecture Notes in Computer Science, vol 8351. Springer, Cham
- [16] E. M. Sipayung, H. Maharani, and B. A. Paskhadira, "Designing Customer Target Recommendation System Using K-Means Clustering Method," in IJITEE, Vol 1, No. 1, March 2017, pp 1-7.
- [17] D. Sarma1, T. Mitra and M.S. Hossain,"Personalized Book Recommendation System using Machine Learning Algorithm," in (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 12, No. 1, 2021
- [18] S. Vimala and K. Vivekanandan, "A Kullback–Leibler divergence-based fuzzy C-means clustering for enhancing the potential of an movie recommendation system," SN Applied Sciences, vol. 1, no. 7, p. 698, 2019.
- [19] Shouval, R., Bondi, O., Mishan, H. et al. Application of machine learning algorithms for clinical predictive modeling: a data-mining approach in SCT. Bone Marrow Transplant 49, 332–337 (2014). <https://doi.org/10.1038/bmt.2013.146>